Lawson OLIVEIRA LIMA

AI/ML Engineer

Male linkedin.com/in/lawson-oliveira-lima lawson.olima@gmail.com

AI/ML Engineer with a Master's degree from CentraleSupélec (top French engineering school) and a Bachelor's degree from UFC (Brazil). Specialized in Software Engineering, Geometric Deep Learning and GNNs, with experience in Time Series, Anomaly Detection, Audio Processing and Agentic AI (RAG, Multi-Agent Orchestration, LLMOps).

EDUCATION

2021 – 2025 **Master's in Engineering — GPA: 3,79/4** CentraleSupélec – Paris-Saclay, France

Scholarship: *CentraleSupélec Foundation Excellence Scholarship*

Relevant Courses: AI/ML, Bayesian Learning, Signal and Image Processing, Control Theory, Optimization, Algorithms and Complexity, Measure Theory, and Probability.

2019 – 2025 **Electrical Engineering — GPA: 9,47/10** Federal University of Ceará – Fortaleza, Brazil

Distinction: *Summa Cum Laude (Highest academic honor awarded to the top-ranking student)*.

Relevant Courses: Statistics, C Programming, Computer Architecture, Analog and Digital Electronics, Communication Principles.

PROFESSIONAL EXPERIENCE

07/24 – Present **AI Engineer (Backend & Infra)** Casa dos Ventos (TotalEnergies Joint Venture) – Brazil

- Architected a multi-agent system with guardrails, integrating WhatsApp, Salesforce Service Cloud, MCP servers and Langfuse for production observability and security regression testing.
- Designed an agentic RAG layer connecting LLM agents to PostgreSQL via MCP, secured with Row-Level Security (RLS) for per-user/role data scoping.
- Built OCR pipelines for document extraction, reaching approximately 100 weekly active users.
- Own DevSecOps for the company's internal software systems, using AI to continuously track vulnerabilities and propose effective remediations.

RESEARCH EXPERIENCE

06/23 – 01/24 **Machine Learning Researcher (Intern)** Université Paris-Saclay & Transvalor – France

Lab: MICS (Mathematics and Informatics for Complexity and Systems)

- Designed a Geometric Deep Learning framework using PyTorch Geometric to emulate metal forging physics.
- Managed large-scale geometric datasets (∼1TB) and orchestrated distributed training on HPC clusters using Slurm and Multi GPU (4xA100).

PUBLICATIONS & ACADEMIC THESES

- **L. O. Lima**. "A Hybrid GNN-Heuristic Approach for Unsupervised Partitioning of Featureless Graphs." *Undergraduate Thesis - CentraleSupélec & Federal University of Ceará, Brazil, 2025*. [Link].
- **L. O. Lima**, et al. "Multilayer Perceptron-based Surrogate Models for Finite Element Analysis." *Conference Paper at DCABES - Chizhou University, China, 2022*. [Link]

SKILLS

Programming	Python, Rust, C++, TypeScript, NodeJS, Matlab, SQL
Background	Graph Theory, Linear Algebra, Optimization, Statistics, Audio/Image Processing
Frameworks	PyTorch, Jax, Lightning, Pytorch Geometric, Wandb, Pandas, Matplotlib, Flask
GenAI/LLMOps	Multi-Agent Systems, MCP, LiteLLM, Langfuse, RAG, Guardrails, CI/CD, Google ADK
Infra	Terraform, Slurm, Docker, Google Cloud, Serverless Systems
Languages	Portuguese (Native), English (C1), French (C1), Spanish (A1)

AWARDS

• **Summa Cum Laude (UFC)** • **CentraleSupélec Foundation Excellence Scholarship** • **Google Cloud Associate**